

# CHEMISTRY | UNIT 03 | LESSON 05

## Intermolecular Forces

### Checkpoint Answer Sheet

Name: \_\_\_\_\_ Date: \_\_\_\_\_

**MLA standards:** MLA.CHEM.BON.01

Use the data and complete the checkpoint independently.

| Substance        | Shape / polarity clue | Boiling point (C) | Strongest IMF |
|------------------|-----------------------|-------------------|---------------|
| CH <sub>4</sub>  | Nonpolar, symmetric   | -161.5            |               |
| NH <sub>3</sub>  | Polar; N-H bonds      | -33.3             |               |
| H <sub>2</sub> O | Polar; O-H bonds      | 100.0             |               |

1. Rank CH<sub>4</sub>, NH<sub>3</sub>, and H<sub>2</sub>O from weakest to strongest overall intermolecular attraction.

2. Explain the boiling-point pattern using intermolecular forces.

3. Draw two water molecules. Label partial charges and the hydrogen-bond attraction between molecules.

4. CER: Which substance has the strongest intermolecular attraction? Use two data points.